What do I do with my data?

Visualizing a transparent author workflow for data sharing and open research publishing

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Background

Data sharing is an important part of open research, but it can be a challenge for authors, particularly in subject areas where data sharing is not already established practice. Many academic publishers have introduced editorial policies which suggest or mandate open data, but these text-heavy pages can be complex and opaque, often assuming significant experience with open data practice.

F1000 has always advocated open research, and we have a particularly 'open' policy on data sharing. We have developed this novel, visual workflow to help authors navigate the process of identifying, preparing and describing their data ahead of article submission. Our editorial teams also use the workflow to check that an author has shared the correct data in the recommended way.

With a view to reaching authors across the disciplines and around the world, the workflow is designed to make it easier for authors to publish in open research journals by clarifying general guidelines and specific submission requirements. The aim is to distil a complex data policy into a set of practical steps displayed in a visually engaging way. This innovative tool improves transparency in two ways: by clarifying publisher expectations around data sharing and by facilitating the open sharing of data by researchers.

Approach

The workflow focuses on the author's point of view, beginning with the question of whether or not they have data to share. The goal is to welcome and support authors who are unfamiliar with open data, and to save time for more experienced authors who want to double check submission requirements. The workflow then guides authors through identifying what kind of data should be shared, and thereby clarifies what editors are looking for. Rather than assigning an interdisciplinary authorship to arbitrary subject categories, authors are invited to follow the pathway most reflective of their research approach.

> *Some may say there is no better motivator for a researcher to share their data than if a publication is at stake. When asked who they would be willing to receive support from, the most popular answer was publishers (41%) closely followed by those within their own institution (38%).*[1]
>
> Mark Hahnel *The State of Open Data 2022*

Conclusions

The initial stages of the workflow identify what kinds of data are used and expected with a given article type and research design – reported by our editorial teams as a knowledge gap for many of our authors. In building this resource to address this gap, we learned valuable lessons which we will apply to developing the rest of the workflow: the importance of consulting both internal and external stakeholders, the need to represent a range of experience levels and disciplinary viewpoints, and the challenges of creating simple graphic models of complex decision-making and working practices.

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References

[1] Hahnel, M. (2022) 'Foreword', in Digital Science (2022) *The State of Open Data 2022*. (Online). Available from: doi:10.6084/m9.figshare.21276984.v5 (Accessed: 13 February 2023).

"Will I be expected to share data? I'm submitting a...

Case report	Review	Research article	Data note	Policy brief
Correspondence article	Study protocol	Brief report	Genome note	Registered report
Editorial		Case study	Living systematic review	Software tool article
Opinion article		Clinical practice article	Method article	Systematic review

We would not typically expect you to share new data in a data repository for these article types. You can contact our editorial team to discuss your article if you have data to share or cite.

These article types may have associated data that we would expect to be shared in an appropriate data repository. The next steps in the workflow will help you to identify your data, whether it was generated for your study or reused from another source.

"What kind of data are you expecting with my article? My approach to this project was...

...simulation	...experimental	...social	...inventive	...analytical
Computer Models, Artificial Intelligence / Machine Learning, Video Games, Molecular Modelling	Synthesis, Materials Testing, Cell Studies, Clinical Trials, Microscopy, Survey, Genetic Sequencing, Bibliometrics, Content Analysis	Survey, Interview, Focus Group, Annotation, Field Research, Action Research, Oral History	Robotics, Equipment Design, Curriculum Development, Film Production, Game Design, Exhibition or Collection Development, 3D Printing	Meta-analysis, Bibliometrics, Textual Analysis, Replication Studies, Data Reuse and Reanalysis
Software, Code, Spreadsheets, Images, Videos, Audio Files, CAD Files	Spreadsheets, Spectra, Crystal Structures, Cell Images, Genetic Sequences, Statistics, Survey Responses, Lists, Census Data, Text Files	Survey Free Responses, Interview Transcripts or Recordings, Annotations, Field Notes, Sketches, Photos, Diaries, Text Files	Software, Code, Images, Audio & Video Recordings, 3D Printing Files, Digital Archive, Exhibition, Transcripts, Text Files, Presentation Slides	Published Sources, Archival Sources, Maps, Reference Lists, Gene Sequences, Digital Output Files from Pre-existing Dataset(s)

Example data to share or cite

Now that you have identified the data we would expect with your article, **we can help you prepare it for sharing and publication**.

This is your data. We will need you to describe how you accessed it, but you don't need to deposit it into a data repository.

"How do I prepare and share my data appropriately?

Next steps

After data identification, the workflow moves on to other reported challenges for authors, including how to prepare research data, share it in a repository, describe it accurately, and cite it in an article.

This is a preview of our Preparing Your Data Workflow.

Can you tell me about...

FAIR	FAIR	FAIR	FAIR
• Making data FINDABLE • Data sharing standards • Describing data / adding metadata • Linking to datasets in citations	• Making data ACCESSIBLE • Choosing a repository • Uploading datasets • Applying an open licence to your data	• Making data INTEROPERABLE • Formatting data files • Open access software	• Making data REUSABLE • Disaggregating data for sex & gender equity • Anonymizing personal data